

Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

Chapter 05 The Working Cell



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Some Basic Energy Concepts

- **Energy** is defined as the capacity to cause change.
- Some forms of energy are used to perform work, such as moving an object against an opposing force.

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Conservation of Energy

- **Kinetic energy** is the energy of motion.
- **Potential energy** is stored energy, energy that an object has because of its location or structure.
- Life depends on countless similar conversions of energy from one form to another.



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Conservation of Energy

- A physical principle known as **conservation of energy** explains that it is not possible to destroy or create energy.
- Energy can only be converted from one form to another.

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Identifying Major Themes

You turn the crank of a well, raising a bucket of water from below ground to the surface.

Which major theme is illustrated by this action?

1. The relationship of structure to function
2. Information flow
3. Pathways that transform energy and matter
4. Interactions within biological systems
5. Evolution

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Heat

- **Heat** is a type of kinetic energy contained in the random motion of atoms and molecules.
- All energy conversions generate some heat.
- **Entropy** is a measure of disorder, or randomness, in a system.
- Every time energy is converted from one form to another, entropy increases.

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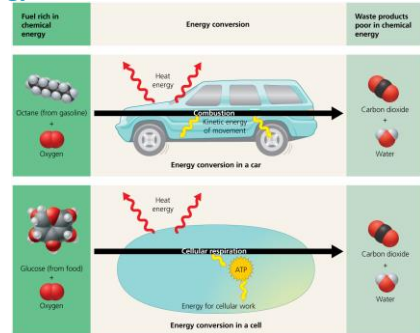
Chemical Energy

- The molecules of food, gasoline, and other fuels have a form of potential energy called **chemical energy**, which arises from the arrangement of atoms and can be released by a chemical reaction.
- Living cells and automobile engines use the same basic process to make the chemical energy stored in their fuels available for work.

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Energy Transformations in a Car and a Cell



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Chemical Energy

- Cellular respiration** is
 - the energy-releasing chemical breakdown of fuel molecules and
 - the storage of that energy in a form the cell can use to perform work.
- Humans convert about 34% of our food energy to useful work. The rest of the energy released by the breakdown of fuel molecules generates body heat.

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Food Calories

- A **calorie** (cal) is the amount of energy that can raise the temperature of 1 gram (g) of water by 1°C.
- Food calories are kilocalories, equal to 1,000 calories.
- The energy of calories in food is used to fuel the activities of life.

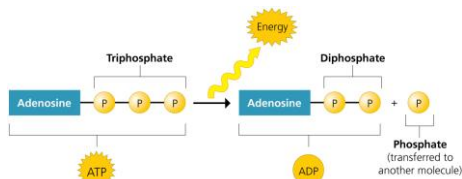


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ATP and Cellular Work

- Chemical energy released by the breakdown of organic molecules during cellular respiration is used to generate molecules of ATP.
- ATP** (adenosine triphosphate)
 - acts like an energy shuttle,
 - stores energy obtained from food, and
 - releases it later as needed.



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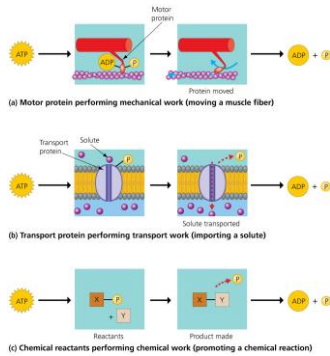
Phosphate Transfer

- ATP energizes other molecules in cells by transferring phosphate groups to those molecules.
- This energy
 - helps cells change shape,
 - enables the transport of ions and other dissolved substances across the membranes, and
 - drives the production of a cell's large molecules.

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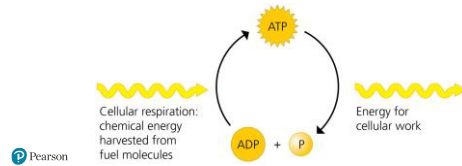
How ATP Drives Cellular Work



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The ATP Cycle

- Cells spend ATP continuously.
- Cellular work spends ATP, which is recycled when ADP and phosphate are combined, using energy released by cellular respiration.
- Up to 10 million ATPs are consumed and recycled each second in a working muscle cell.



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Enzymes

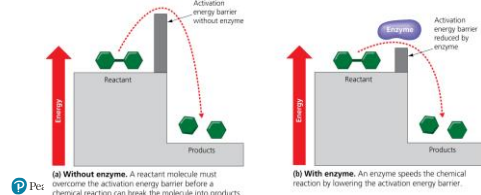
- Metabolism** is the total of all chemical reactions in an organism.
- Most metabolic reactions require the assistance of **enzymes**, proteins that speed up chemical reactions without being consumed by the reaction.
- All living cells contain thousands of different enzymes, each promoting a different chemical reaction.

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Activation Energy

- Activation energy** is the energy that must be invested to start a reaction, by This initial split requires an energy "investment" of two ATP molecules per glucose.
 - activating the reactants and
 - triggering a chemical reaction.
- Enzymes enable metabolism to occur by reducing the amount of activation energy required to break the bonds of reactant molecules.



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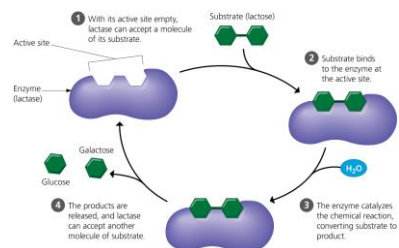
Structure/Function: Enzyme Activity

- An enzyme is very selective in the reaction it catalyzes.
- Each enzyme recognizes a **substrate**, a certain reactant molecule.
 - The **active site** has a shape and chemistry that fits the substrate molecule.
 - This interaction is called **induced fit**, because the entry of the substrate induces the enzyme to change shape slightly, making the fit between the substrate and active site snugger.
- After the products are released from the active site, the enzyme can accept another molecule of its substrate.
- The ability to function repeatedly is a key characteristic of enzymes.
- Many enzymes are named for their substrates, but with an **-ase** ending.

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How an Enzyme Works



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Structure/Function: Enzyme Activity

Major Themes: Enzymes unravel and stop functioning if the environment gets too hot.

Which major theme is illustrated by this action?

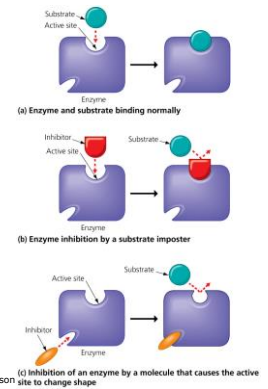
1. The relationship of structure to function
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Enzyme Inhibitors

- Certain molecules inhibit a metabolic reaction by
 - binding to an enzyme and
 - disrupting its function.
- Some of these **enzyme inhibitors** are actually substrate imposters that plug up the active site.
- Other inhibitor disrupts the function of an enzyme by altering its shape.



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Identifying Major Themes

Your ability to walk depends upon the coordination of many different enzymes and other cellular structures.

Which major theme is illustrated by this action?

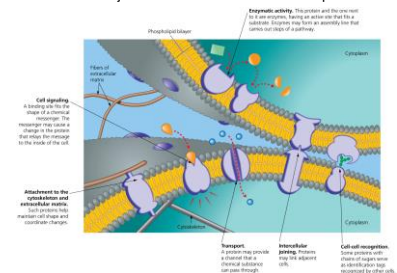
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Membrane Function

- Cells must also regulate the flow of materials to and from the environment.
- The plasma membrane consists of a double layer of fat (a phospholipid bilayer) with embedded proteins.
- **Figure 5.11** describes the major functions of these membrane proteins.

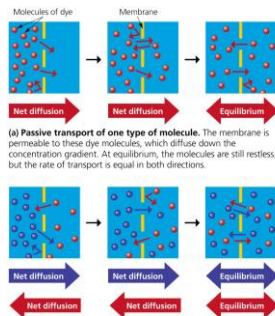


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Passive Transport: Diffusion Across Membranes

- Molecules constantly vibrate and wander randomly.
- **Diffusion** is the movement of molecules to spread out evenly into the available space.
- **Passive transport** is the diffusion of a substance across a membrane without the input of energy.
- In passive transport, a substance diffuses down its **concentration gradient** from where the substance is more concentrated to where it is less concentrated.



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Passive Transport: Diffusion Across Membranes

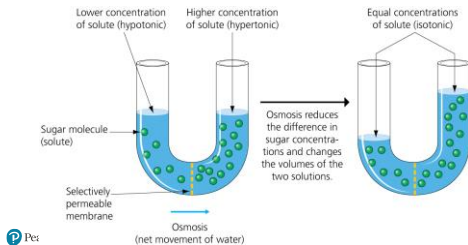
- Substances that do not cross membranes spontaneously, or otherwise cross very slowly, can be transported via proteins that act as corridors for specific molecules.
- This assisted transport is called **facilitated diffusion**, a type of passive transport.

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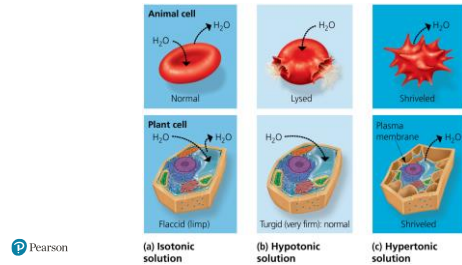
Osmosis and Water Balance

- The diffusion of water across a selectively permeable membrane is **osmosis**.
- A **solute** is a substance that is dissolved in a liquid solvent, such as the salt in salt water, and the resulting mixture is called a solution.



Osmosis and Water Balance

- Compared to another solution,
 - a **hypertonic** solution has a higher concentration of solute,
 - a **hypotonic** solution has a lower concentration of solute, and
 - an **isotonic** solution has an equal concentration of solute.



Water Balance in Animal Cells

- The survival of a cell depends on its ability to balance water uptake and loss.
- For an animal to survive a hypotonic or hypertonic environment, the animal must have a way to balance the uptake and loss of water.
- The control of water balance is called **osmoregulation**.

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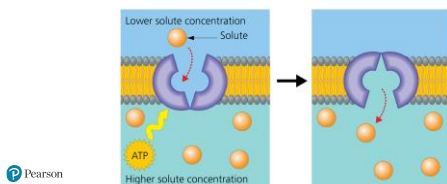
Water Balance in Plant Cells

- Plant cells have rigid cell walls.
- Plant cells are healthiest in a hypotonic environment with a net inflow of water, which expands their cell walls without bursting.



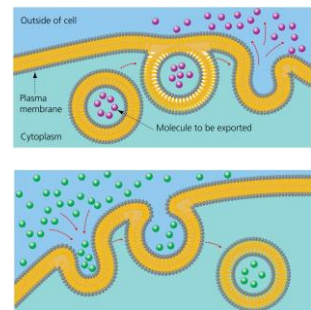
Active Transport: The Pumping of Molecules Across Membranes

- Active transport** requires that a cell expend energy to move molecules across a membrane.
 - Cellular energy (usually provided by ATP) is used to drive a transport protein that pumps a solute *against* the concentration gradient.
 - Active transport allows cells to maintain internal concentrations of small solutes that differ from environmental concentration.



Exocytosis and Endocytosis: Traffic of Large Molecules

- Exocytosis** is the movement of materials out of the cytoplasm of a cell via membranous vesicles or vacuoles that fuse with the plasma membrane.
- In **endocytosis**, a cell takes material in via vesicles that bud inward. For example, in a process called **phagocytosis** ("cellular eating"), a cell engulfs a particle and packages it within a food vacuole.



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